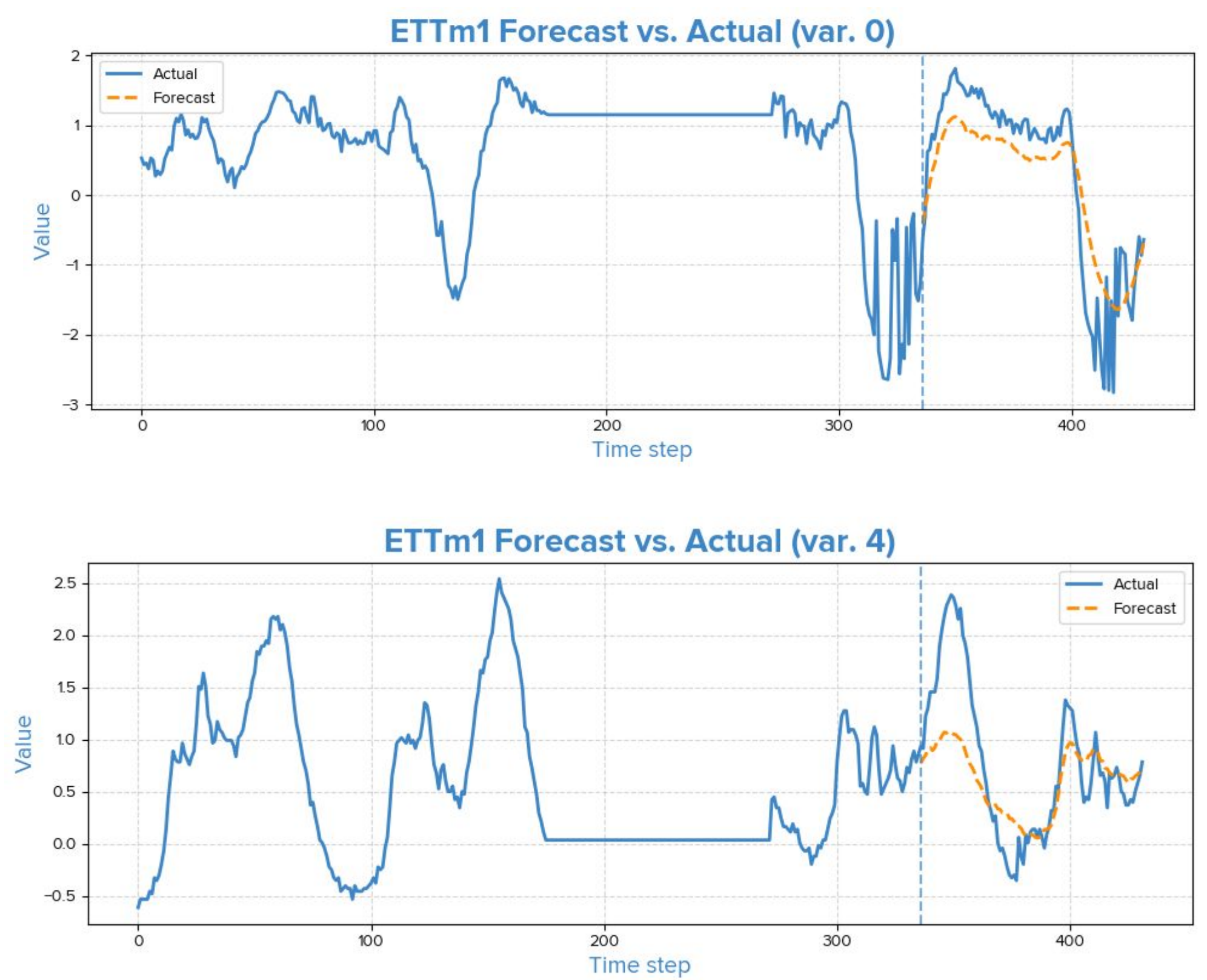


A Time Series is Worth 64 Words: Long-term Forecasting with Transformers

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Background

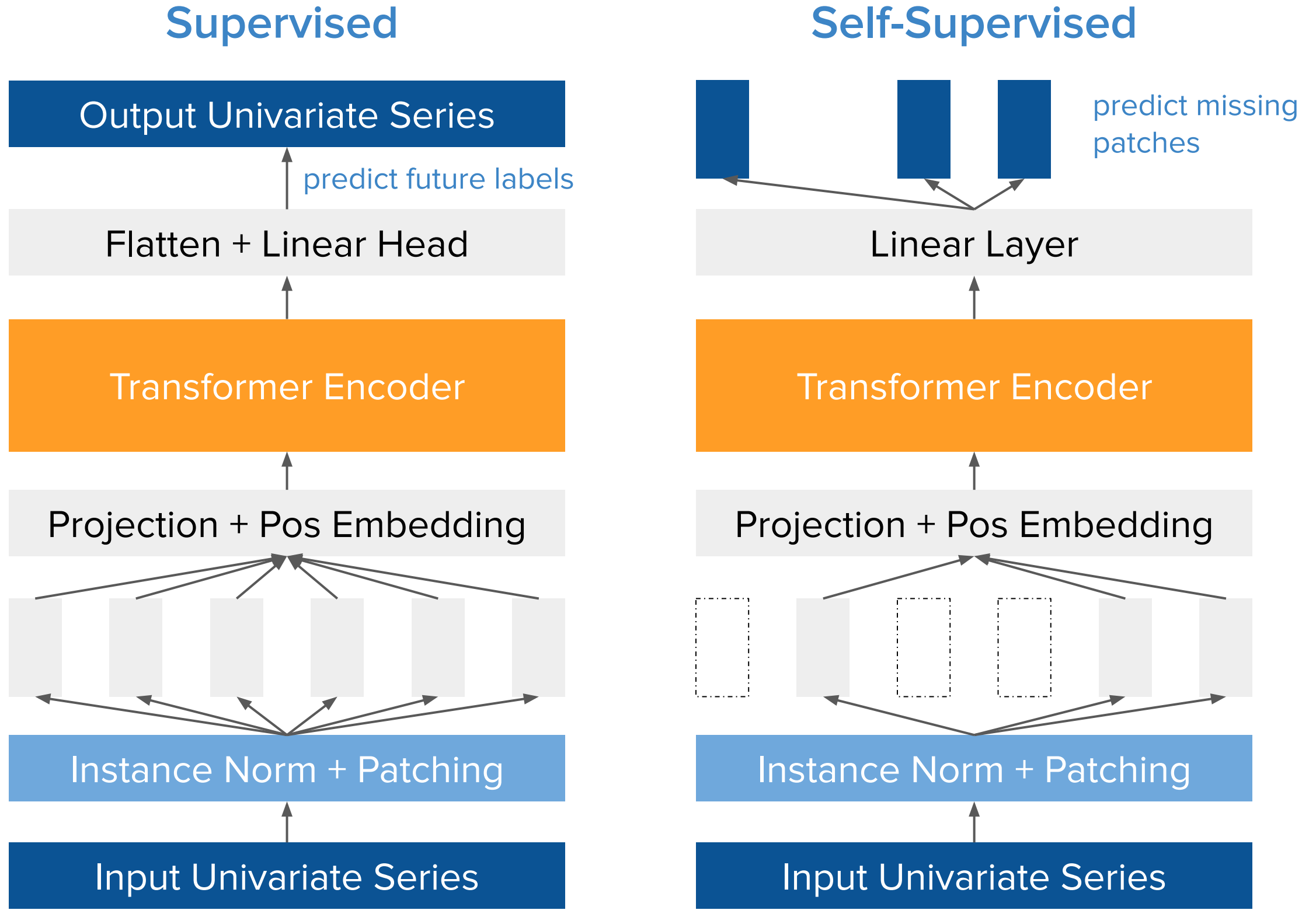
- **Long-term forecasting** of multivariate time series is critical in domains like energy, traffic, and finance
- Transformer-based models often **struggle** to balance forecasting **accuracy** with computational **efficiency**
- Project goals: reproduce PatchTST’s superior forecasting accuracy and assess its self-supervised representation learning through **Patching** and **Channel Independence**
- Key benefits: **lower computational cost** and the ability to learn effectively from **longer historical windows**



Key Ideas

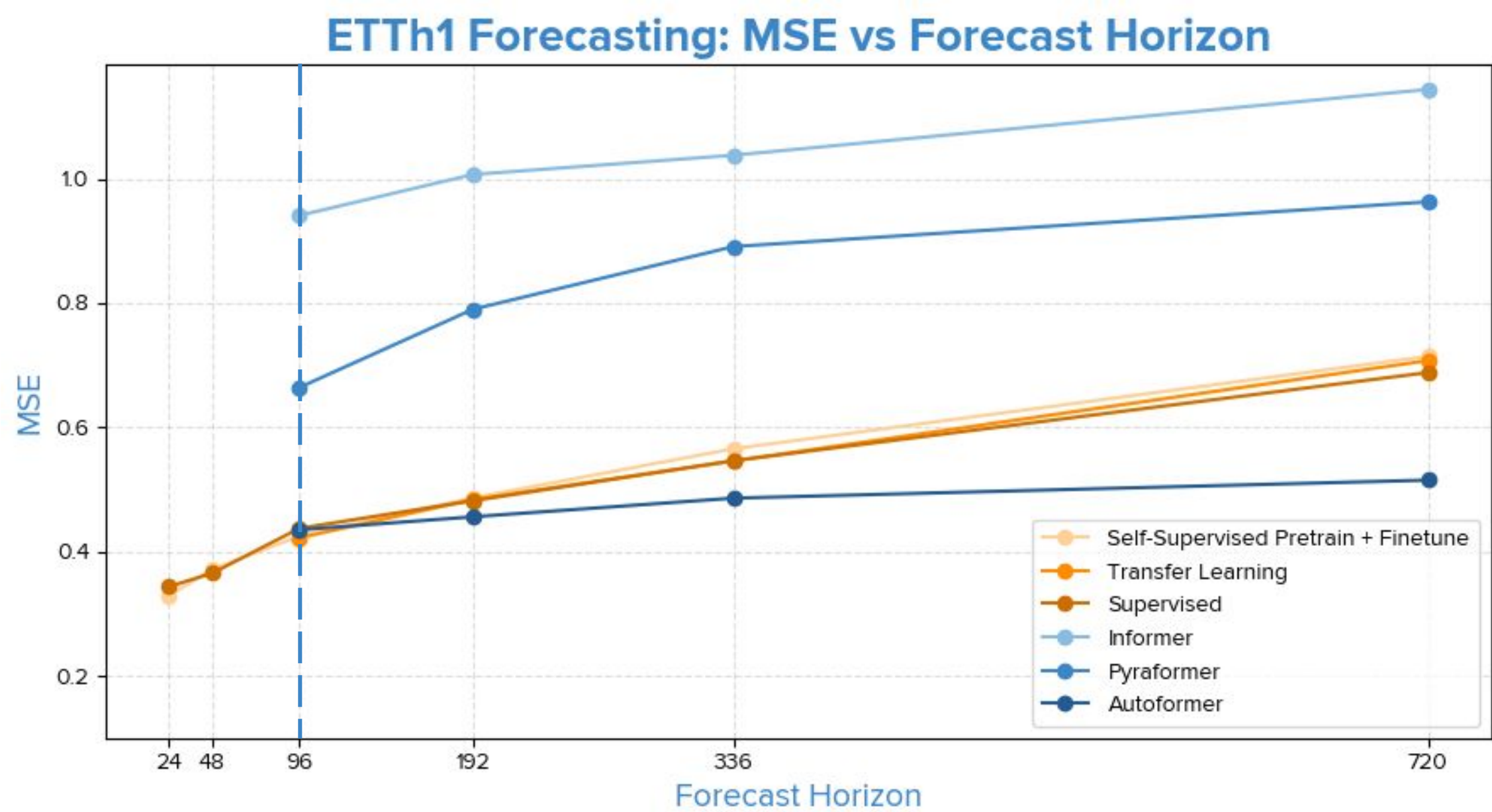
- Patching:** Divide each multivariate time-series window of length L into patches of length p with stride s
- Channel Independence:** Divide multivariate time series data into different channels—each channel still share the same Transformer backbone
- Instance Norm:** Normalize each time series instance with zero mean and unit standard deviation before patching
- Transformer Encoder:** Use a vanilla transformer encoder layer & perform multi-head attention

Methodology



Datasets: Two electricity transformer datasets–ETTh1 & ETTm1, one electricity consumption data used for transfer learning

Results and Evaluation



Using a 96 unit forecast horizon: **34%** reduction in MSE compared to Pyraformer, **54%** compared to Informer and comparable to Autoformer

Dataset	Forecast Horizon	Our PatchTST		Original PatchTST	
		MSE	MAE	MSE	MAE
ETTh1	96	0.4373	0.4638	0.375	0.3999
	192	0.4818	0.4921	0.414	0.421
	336	0.5463	0.5334	0.431	0.436
ETTM1	96	0.3352	0.3808	0.290	0.342
	192	0.3725	0.4093	0.332	0.369
	336	0.4177	0.4447	0.366	0.392

Conclusions

- Patching & channel independence gave better results & faster training time than most other time series transformers
- We were not able to find significant differences between supervised and finetuned self-supervised learning
- **Transfer learning** with the Electricity dataset gave us **similar results** with **less epochs** relative to supervised learning
- We had slightly higher loss compared to the original paper, likely due to **implementation differences** in the encoder
- Loss is lower on shorter **forecast horizons**
- We also noticed **no significant difference** in loss due to changing the patch size, stride, or lookback window

Future Considerations

- Implementing learnable positional encodings
- Trying BatchNorm layers as they tend to have better results with Time Series Transformers
- Apply PatchTST to other datasets, such as finance, health, and environmental data

References

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). A time series is worth 64 words: Long-term forecasting with transformers. arXiv. <https://arxiv.org/abs/2211.14730>